

ABB Ltd — Medium Voltage Products

REQUEST FOR QUOTATION — RING MAIN UNIT (ABB SafeRing GIS)

RFQ-MV-2026-SR-0112 | Rev B | SafeRing (GIS) | Greenfield Solar Farm Phase 2

Document Ref	RFQ-MV-2026-SR-0112	Revision	B
Date	24 June 2026	Client	Greenfield Energy Consortium Ltd
	Greenfield Solar Farm — 33/11 kV Grid Connection, Phase 2		ABB Ltd, Medium Voltage Division

1. SCOPE OF SUPPLY

The Contractor shall supply a Gas-Insulated Ring Main Unit (RMU) for the Greenfield Solar Farm 33/11 kV Grid Connection project. The product family is **ABB SafeRing (GIS)**. Market coverage required: **All**. The assembly shall comply fully with IEC 62271-200.

2. SYSTEM ELECTRICAL REQUIREMENTS

Parameters below are specified as plain text and map directly to the ABB SafeRing filterSelector API. The API Property Name column shows the exact property name expected by the ABB Sales Configurator.

Parameter	Required Value	Unit	API Property Name (varXxx)
Market	All	—	varMarket_Glo
Product	SafeRing (GIS)	—	varProduct_Glo
Rated Voltage	12 kV	kV	varVoltage_Glo
Short-Circuit Rating	16 kA / 1 s	kA	varShortCircuit_Glo
Busbar Continuous Current	630 A	A	varBBcurrent
Number of Panels	4	—	varPanels
Internal Arc (IAC)	AFL	—	varInternalArc_Glo
Application	With MV metering	—	varMVMetering_Glo
Internal Lighting	Yes	—	varInternalLighting_Glo
Heaters	Yes	—	varHeaters_Glo
Motorized Circuit Breaker	Yes	—	varMotorizedCircuitBreaker_Glo

3. INSULATION LEVEL (IEC 62271-200, Um = 12 kV)

Parameter	Value
Rated voltage (Um)	12 kV
Power frequency withstand voltage	28 kV (1 min)
Lightning impulse withstand (BIL)	75 kV peak
Degree of protection	IP 65 (IEC 60529)
Internal arc classification	AFL — 1 s at 16 kA
Ambient temperature range	-25 °C to +40 °C

PANEL SCHEDULE & PROTECTION RELAY SETTINGS

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4. PANEL SCHEDULE

The RMU shall comprise **4 (four) panels**. The **Type** (varType) and **Configuration** (varConfiguration_Glo) for each panel are shown in the table below and repeated on the Single Line Diagram (Page 3).

Panel	Function	Type (varType)	Configuration (varConfiguration_Glo)	Rating
1	Ring Incomer A	Circuit Breaker (Feeder)	SBC CT VT (REF615 + PM)	12kV/16kA/ 630A
2	Ring Through	LBS	DRC	12kV/16kA/ 630A
3	Ring Incomer B	Circuit Breaker (Feeder)	SBC CT (REF615)	12kV/16kA/ 630A
4	Outgoing Feeder	Circuit Breaker (Feeder)	SBC CT (REF615)	12kV/16kA/ 630A

5. INSTRUMENT TRANSFORMER SPECIFICATION

Panel	Device	Ratio	Class	Burden	Purpose
1	CT (prot.)	200/1 A	5P20	15 VA	50/51, 51N protection
1	CT (meter.)	200/1 A	0.5S	10 VA	MV energy metering
1	VT	12/ $\sqrt{3}$ –0.1/ $\sqrt{3}$ kV	0.2	25 VA	PM metering + 27/59
3	CT (prot.)	200/1 A	5P20	15 VA	50/51, 51N protection
4	CT (prot.)	200/1 A	5P20	15 VA	50/51, 51N protection

6. PROTECTION RELAY SETTINGS (ABB REF615)

ANSI	Function	Pick-up	Time Delay	Panels
50/51	Phase Overcurrent	$1.1 \times I_n = 220 \text{ A}$	0.3 s IDMT-EI	1 & 3 & 4
50N/51N	Earth Fault	$0.1 \times I_n = 20 \text{ A}$	0.5 s	1 & 3 & 4
27	Phase Undervoltage	$0.85 \times U_n = 10.2 \text{ kV}$	2.0 s	1
59	Phase Overvoltage	$1.10 \times U_n = 13.2 \text{ kV}$	1.0 s	1
79	Auto-Reclose	1 shot	0.5 s dead time	1 & 3 & 4
86	Lockout / Trip	—	Instantaneous	1 & 3 & 4

7. ANCILLARY EQUIPMENT

The following ancillary equipment SHALL be supplied fitted to all panels of the RMU assembly unless otherwise stated:

a) Internal Arc Pressure Relief. Internal arc classification AFL is required (Section 2). All pressure relief devices shall vent upward away from operator access areas.

b) Internal Lighting. Internal strip lighting shall be fitted to ALL four panels of the RMU. Supply voltage: 230 V AC auxiliary. **Internal lighting required: YES.**

c) Anti-Condensation Heaters. Panel anti-condensation heaters shall be supplied and factory-wired to terminal blocks in each panel. Supply: 230 V AC, thermostatically controlled. **Heaters required: YES — heaters required for all four panels.**

d) Motorized Circuit Breaker (Motor Operator). A motor-operated mechanism shall be fitted to the vacuum circuit breaker in Panels 1, 3 and 4 to enable remote SCADA close/open commands via 110 V DC. **Motorized circuit breaker required: YES.**

e) Padlocking provisions on all earthing switches and panel doors.

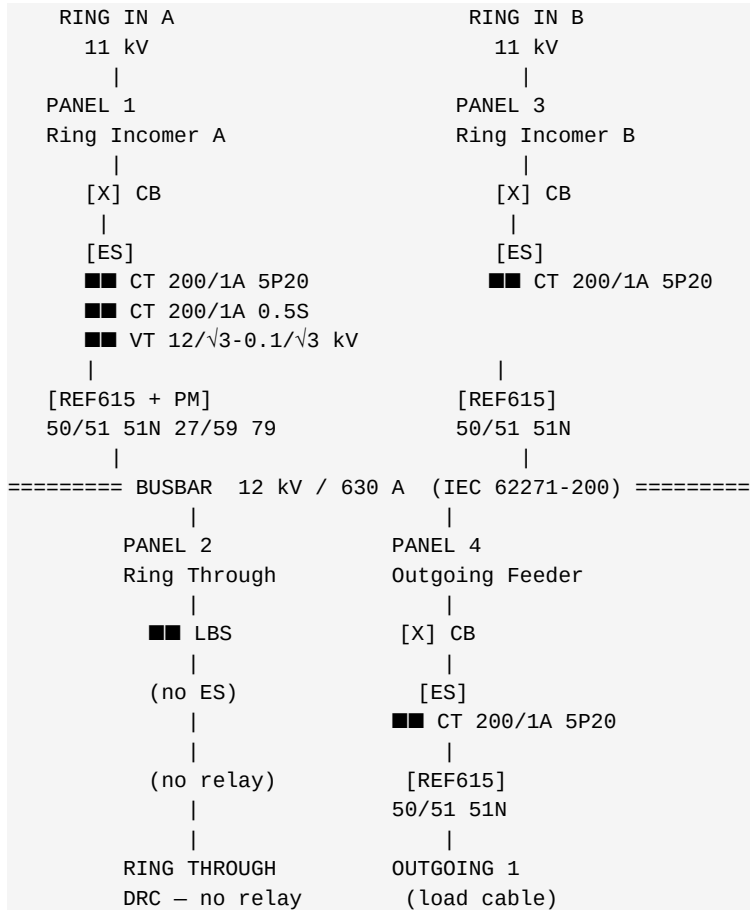
f) Mechanical position indicators — flag type, visible from panel front face.

SINGLE LINE DIAGRAM — SafeRing GIS (4 Panel / 12 kV / 16 kA / 630 A)

Drawing SR-SLD-0112 | Rev B | Not to scale | For Quotation Only

Symbol Legend:

[X] Circuit Breaker (CB)	■ Load Break Switch (LBS)	■ Current Transformer (CT)
■ Voltage Transformer (VT)	[REF615] Protection Relay	■ Earthing Switch (ES)



8. PANEL SUMMARY (Type and Configuration extractable from table below)

Panel	Function	Type	Configuration	Rating
1	Ring Incomer A	Circuit Breaker (Feeder)	SBC CT VT (REF615 + PM)	12kV/16kA/ 630A
2	Ring Through	LBS	DRC	12kV/16kA/ 630A
3	Ring Incomer B	Circuit Breaker (Feeder)	SBC CT (REF615)	12kV/16kA/ 630A
4	Outgoing Feeder	Circuit Breaker (Feeder)	SBC CT (REF615)	12kV/16kA/ 630A

9. RATING SUMMARY

Parameter	Value	API Property Name
Product	SafeRing (GIS)	varProduct_Glo
Market	All	varMarket_Glo
Rated Voltage	12 kV	varVoltage_Glo
Short Circuit	16 kA / 1 s	varShortCircuit_Glo

Parameter	Value	API Property Name
BB Current	630 A	varBBcurrent
Panels	4	varPanels
Internal Arc (IAC)	AFL	varInternalArc_Glo
Application	With MV metering	varMVMetering_Glo
Internal Lighting	Yes	varInternalLighting_Glo
Heaters	Yes	varHeaters_Glo
Motorized CB	Yes	varMotorizedCircuitBreaker_Glo